

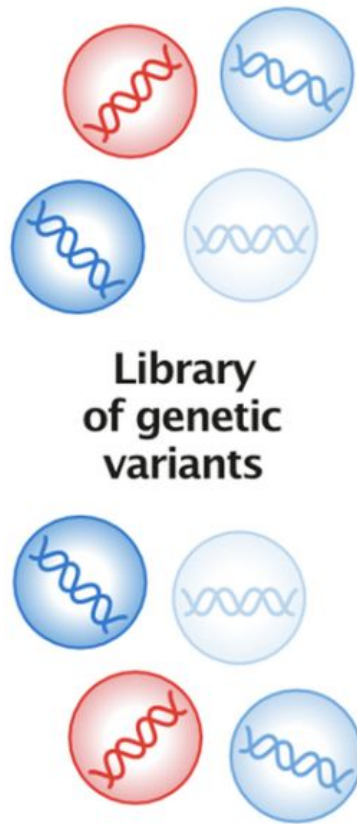
connecting people with science

Dr Alan Rubin

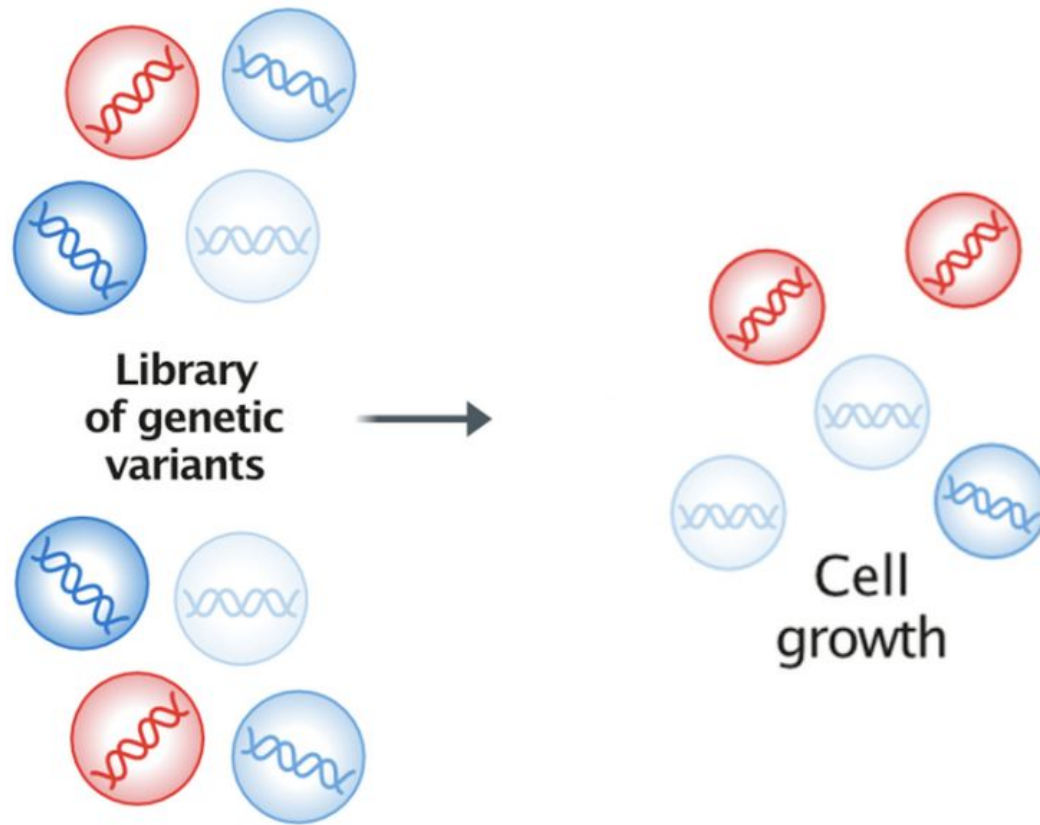
MAVE data analysis III
Data analysis using established software

November 2025

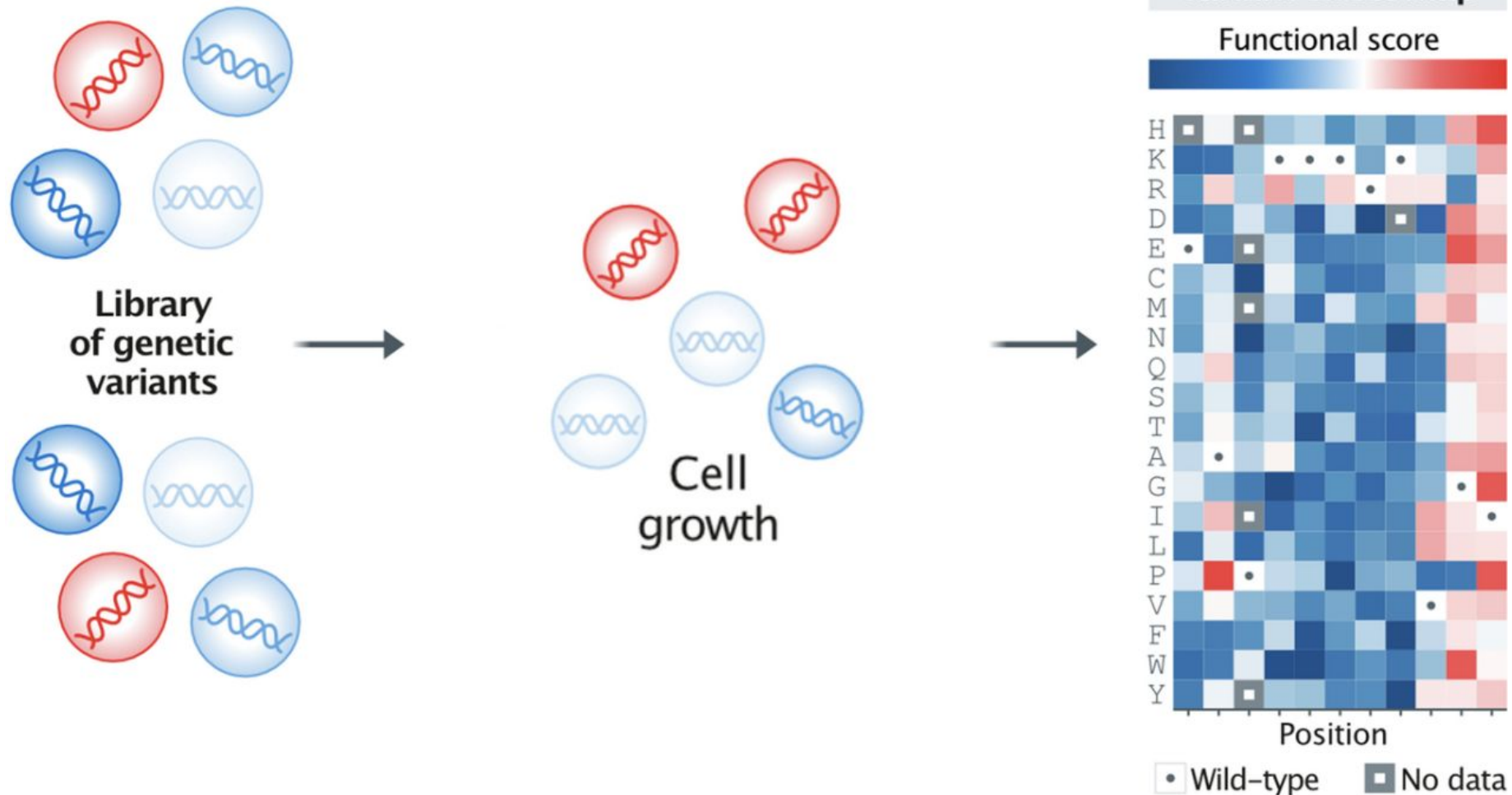
Multiplexed assays of variant effect



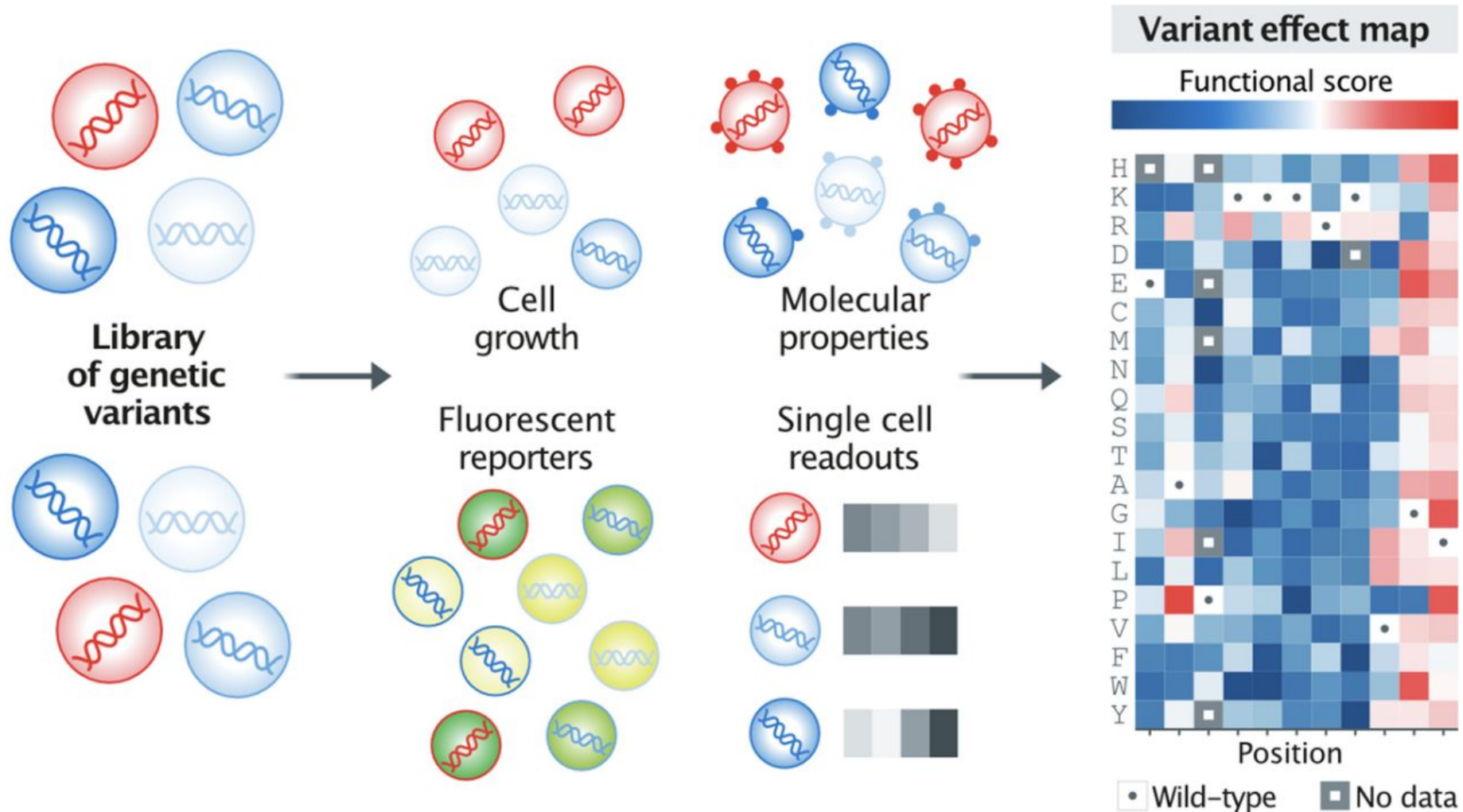
Multiplexed assays of variant effect



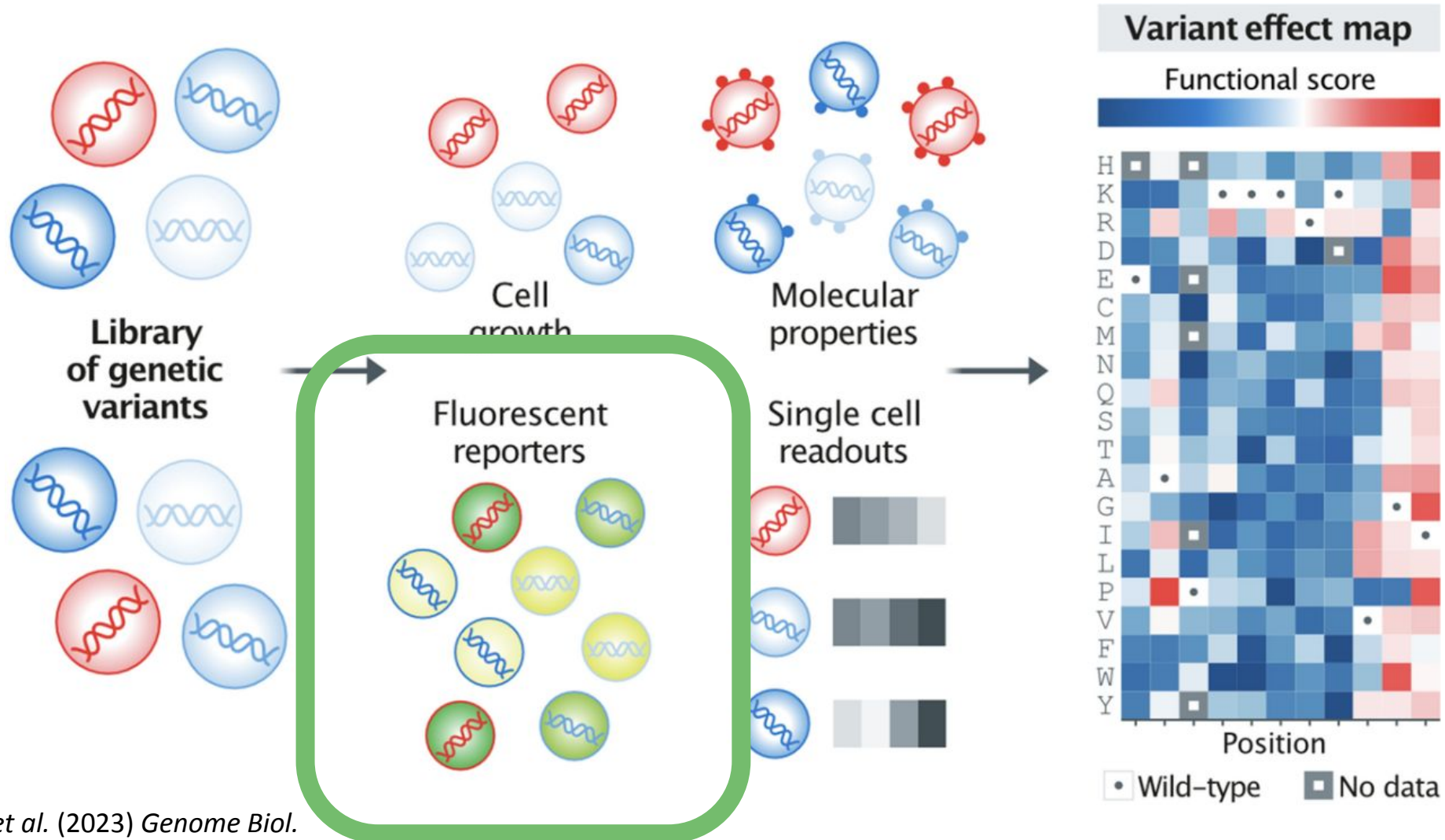
Multiplexed assays of variant effect



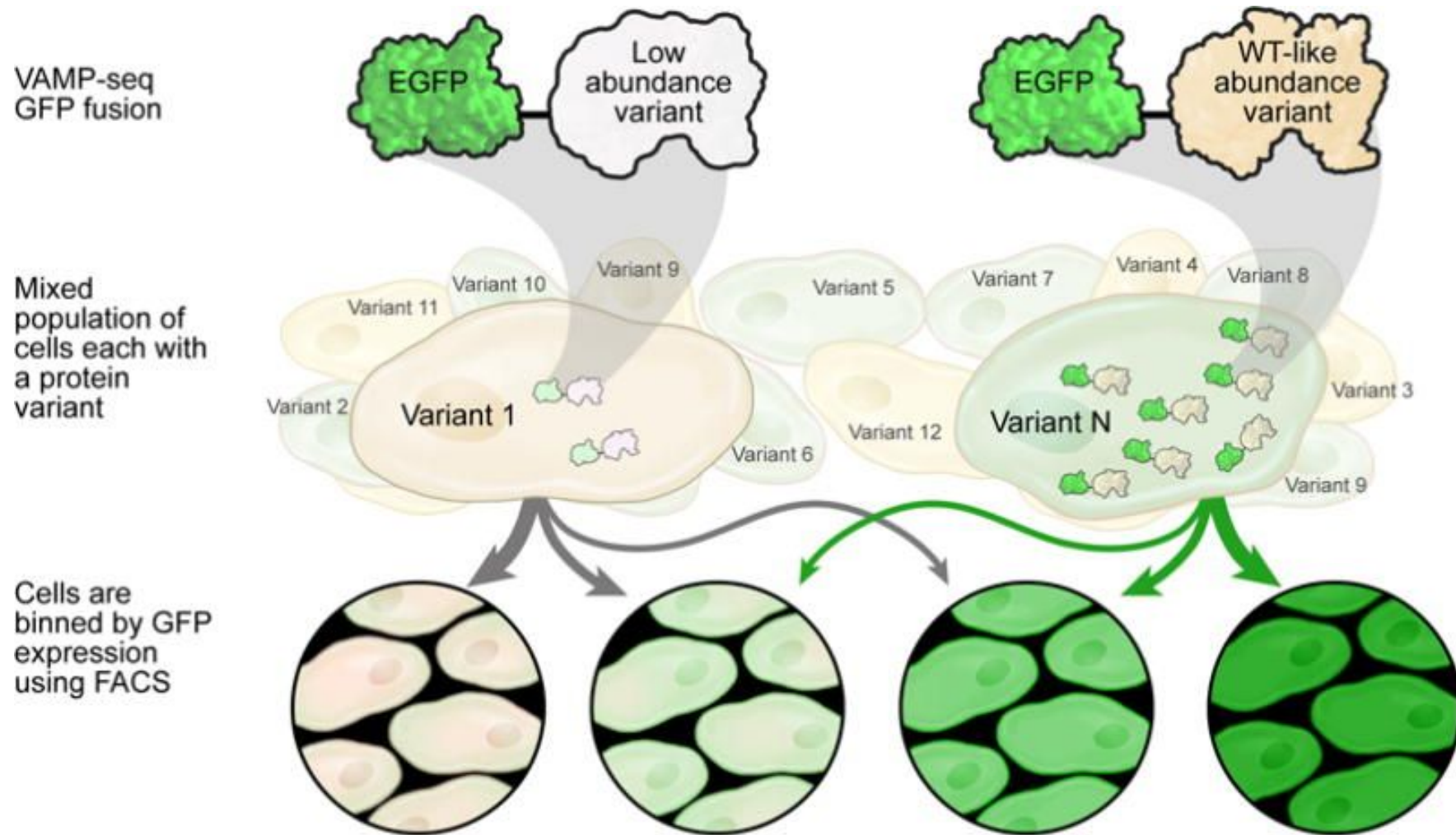
Multiplexed assays of variant effect



Multiplexed assays of variant effect



VAMP-seq: Variant Abundance by Massively Parallel sequencing

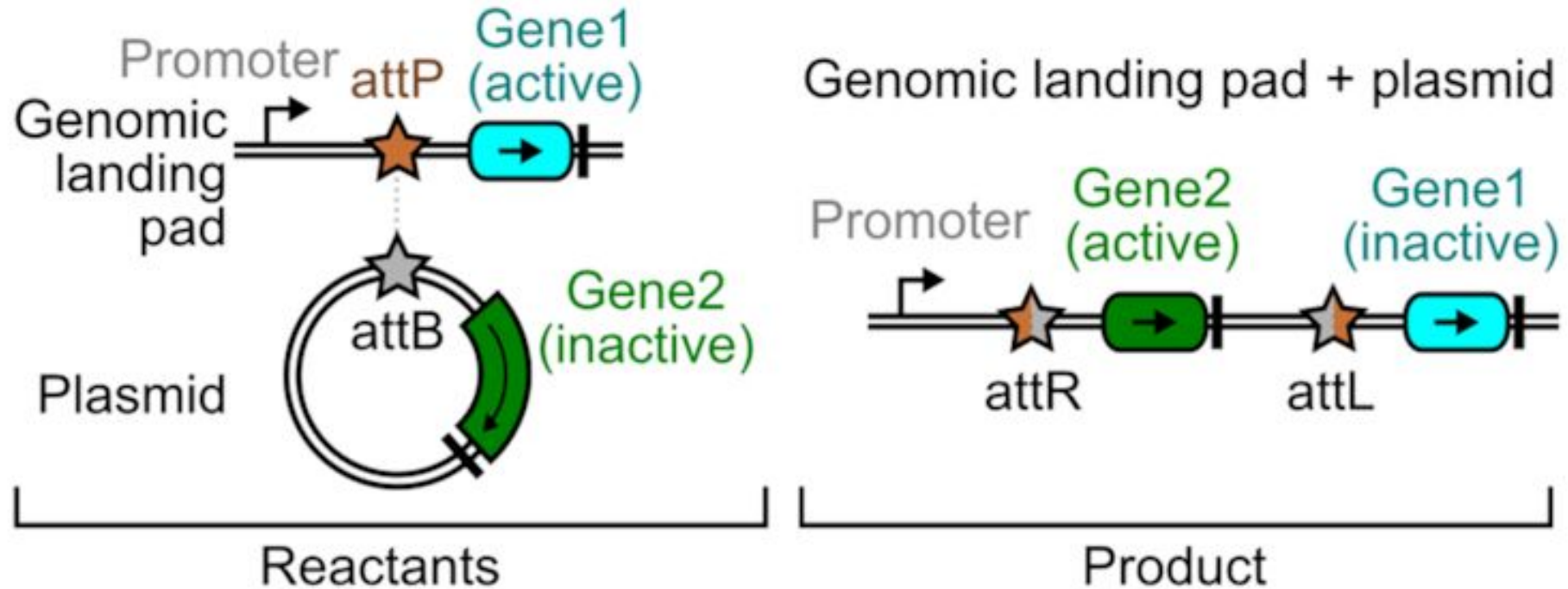


Barcoded cDNA variant libraries



Two sequencing reactions (PacBio and Illumina)

Barcoded cDNA variant libraries



cDNA libraries can be inserted using a landing pad system

VAMP-seq scoring

$$F_{v,total} = \frac{C_{v,bin1} + C_{v,bin2} + C_{v,bin3} + C_{v,bin4}}{\sum C_{bin1} + \sum C_{bin2} + \sum C_{bin3} + \sum C_{bin4}}$$

Step 1: Frequency filtering

VAMP-seq scoring

$$W_v = \frac{(F_{v,bin\ 1} \times 0.25) + (F_{v,bin\ 2} \times 0.5) + (F_{v,bin\ 3} \times 0.75) + (F_{v,bin\ 4} \times 1.0)}{(F_{v,bin\ 1} + F_{v,bin\ 2} + F_{v,bin\ 3} + F_{v,bin\ 4})}$$

Step 1: Frequency filtering

Step 2: Weighted average calculation

VAMP-seq scoring

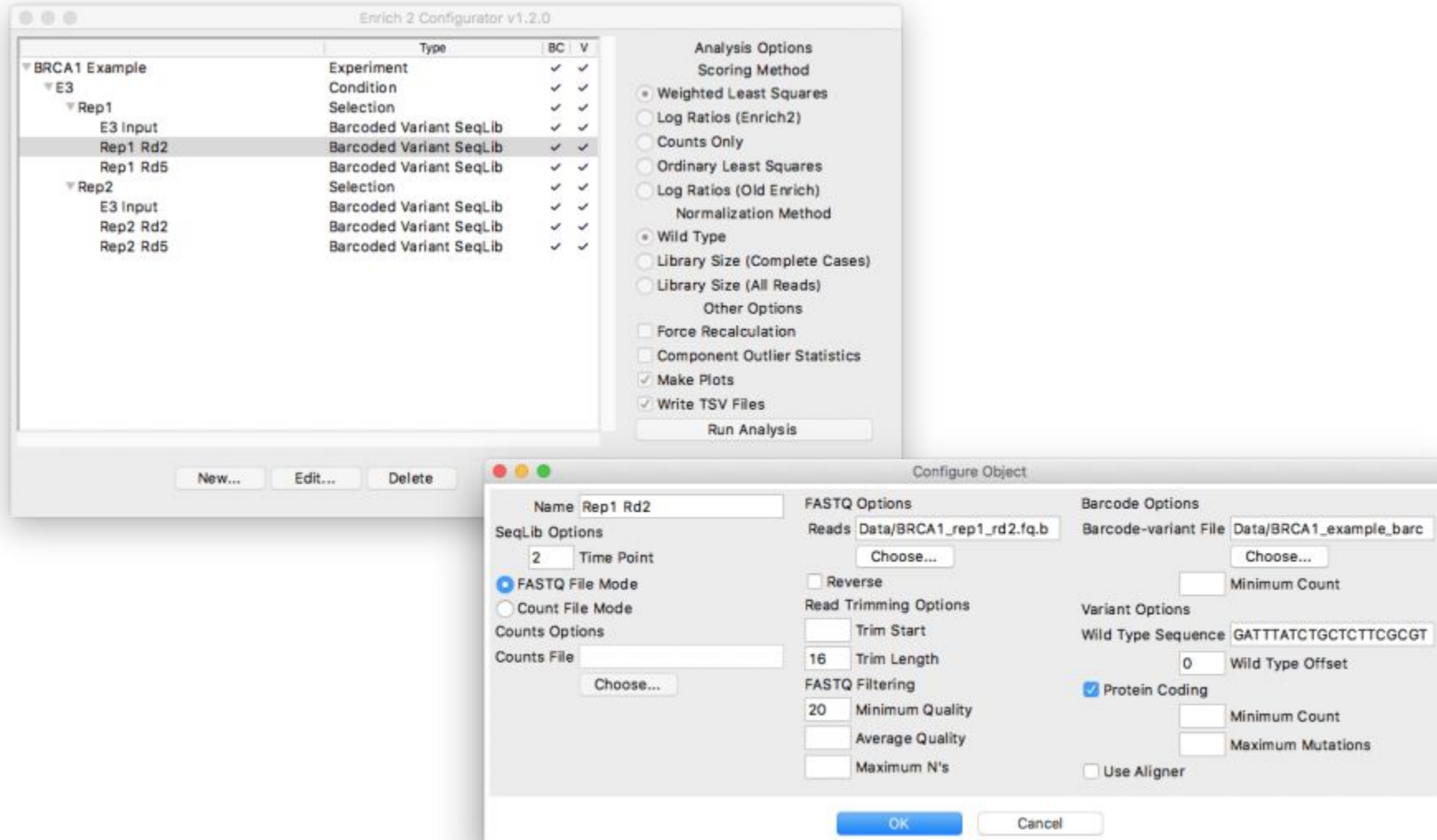
$$S_v = \frac{(W_v - W_{nonsense})}{(W_{wt} - W_{nonsense})}$$

Step 1: Frequency filtering

Step 2: Weighted average calculation

Step 3: Standardisation

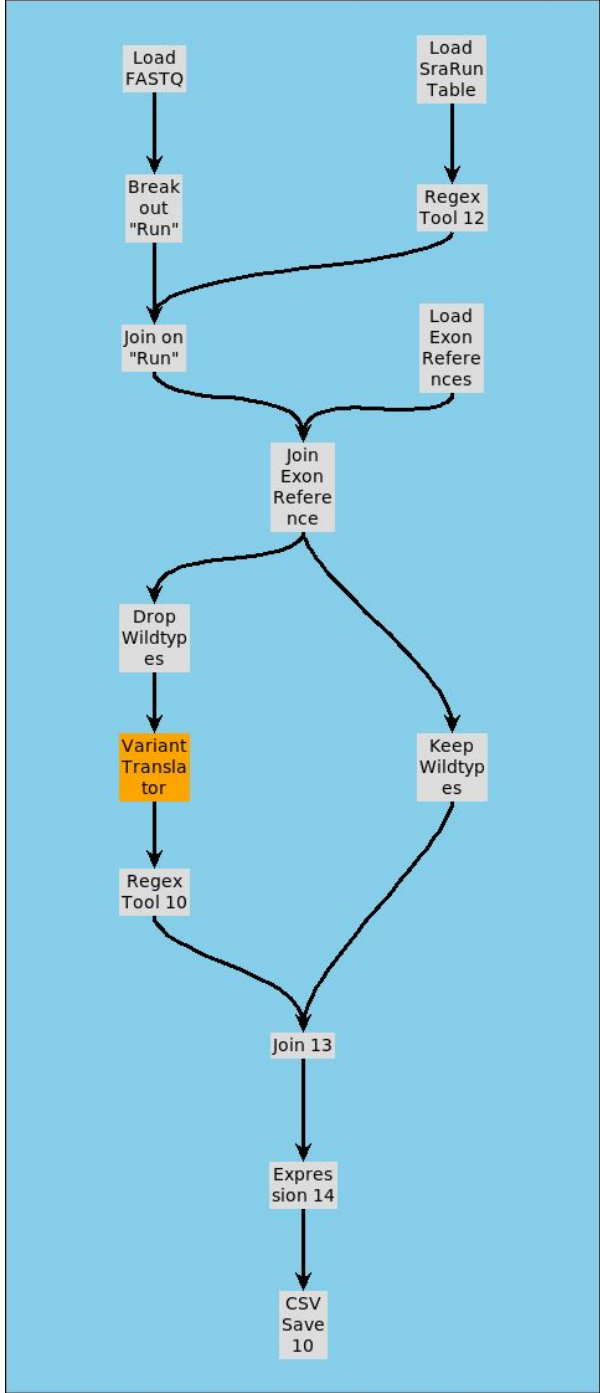
Original VAMP-seq study used Enrich2



Enrich2 benefits and drawbacks

Benefits	Drawbacks
GUI is still easier to use than other tools	Hard to add new statistical methods
Counts direct sequencing and barcode data	Does not support new protocols like SGE and VAMP-seq
Supports coding and noncoding	Python 2.7 support is over

CountESS: Count-based Experiment Scoring and Statistics



Variant Translator

Variant Translator 0.0.33 — Turns a DNA sequence into a HGVS variant code i

[add notes](#)

Input Column

Reference Column

OR Reference Sequence

Automatic Reference Sequence? ☒

Output Column

Max Mutations

Drop unidentified variants ☒

Dataframe Preview 6903 rows

Sample_Name object	exon object	count int64	variant object
X2-lib	2	1	g.65T>C
X2-neg	2	5	g.3T>A
X2-neg	2	1	g.3_5del
X2-neg	2	3	g.6del
X2-neg	2	1	g.8T>A
X2-neg	2	1	g.11del
X2-neg	2	1	g.21T>A
X2-neg	2	2	g.22del
X2-neg	2	6	g.39_40dup
X2-neg	2	1	g.51T>A
X2-neg	2	1	g.65T>G
X2-neg	2	1	g.66T>A
X2-neg	2	1	g.101G>A
X2-neg	2	1	g.127T>C
X2-neg	2	1	g.152C>A
X2-neg	2	1	g.154C>A
X2-neg	2	1	g.158T>A
X2-neg	2	1	g.166C>A
X2-neg	2	1	g.177T>A
X2-neg	2	1	g.179T>A
X2-neg	2	1	g.188G>A
X2-neg	2	1	g.189G>C
X2-neg	2	1	g.207C>A
X2-neg	2	1	g.231C>A

Learning objectives

- Learn how to use DMS analysis software to analyse part of a real dataset
- Work with barcoded cDNA data
- Understand the differences between growth-based data and FACS-based data

Lesson outline

- Brief background lecture
- Running CountESS on an example dataset
 - G6P domain from the G6PD protein
 - Simplified version of VAMP-seq scoring
- Data visualisation

Links and resources

- Data for the example:
https://github.com/WCSCourses/MAVE_2025/tree/main/course_data_2025/VAMP-seq
- CountESS documentation:
<https://countess-project.github.io/CountESS/>
- Bonus content (data visualisation):
https://github.com/WCSCourses/MAVE_2025/blob/main/course_modules_2025/mave_data_analysis_3/VAMP-seq_heatmap.ipynb

